

## 2.4 - Mid of 3.3.

| classical computing                          | quantum computing.                                                               |
|----------------------------------------------|----------------------------------------------------------------------------------|
| classical bits                               | 2-dim quantum sys. described by                                                  |
| $\{0, 1\}$                                   | the qubit space $\mathbb{C}^2$                                                   |
|                                              | state $ \psi\rangle \in \mathbb{C}^2$ .                                          |
|                                              | $\rightarrow$ allow states $a 0\rangle + b 1\rangle$<br>( $ a ^2 +  b ^2 = 1$ ). |
| information storage<br>small                 | big                                                                              |
| Avoiding state changing<br>interactions easy | hard.                                                                            |

### Definition

1. Qubits: quantum mechanical sys. described by 2-dim Hilbert sp.  $\mathbb{C}^2$

2. Qubit space:  $\mathbb{C}^2$ .

$\hookrightarrow$  Select ONB  $\{|0\rangle, |1\rangle\}$ , which consists of eigenstates of  $\hat{\sigma}_z$ .

$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \hat{\sigma}_z |0\rangle = |0\rangle \\ \hat{\sigma}_z |1\rangle = -|1\rangle.$$

### Corollary 2.29.

A measurement of  $\hat{\sigma}_z$  on a qubit yields 1 or -1 as observed val., and projects the qubit in the eigenstate  $|0\rangle$  or  $|1\rangle$  resp.

→ We represent classical bit as a qubit state.

Table 2.1 Representation of classical bits by qubits

| Observed value of $\sigma_z$ | qubit state                        | Represented classical bit value |
|------------------------------|------------------------------------|---------------------------------|
| +1                           | $ 0\rangle =  \uparrow_z\rangle$   | 0                               |
| -1                           | $ 1\rangle =  \downarrow_z\rangle$ | 1                               |

\* Parametrization of pure/mixed qubit state.

1. Pure states.

$$|\psi\rangle \in \mathbb{C}^2.$$

$$\Rightarrow |\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1.$$

$$a = e^{i\alpha} \cos \frac{\theta}{2}, \quad b = e^{i\beta} \sin \frac{\theta}{2}.$$

$$|\psi\rangle = e^{i\alpha} \cos \frac{\theta}{2} |0\rangle + e^{i\beta} \sin \frac{\theta}{2} |1\rangle.$$

$$e^{-i\frac{\alpha+\beta}{2}} |\psi\rangle = e^{i\frac{\alpha-\beta}{2}} \cos \frac{\theta}{2} |0\rangle + e^{-i\frac{\alpha-\beta}{2}} \sin \frac{\theta}{2} |1\rangle$$

$$|\psi(\theta, \phi)\rangle = e^{-i\frac{\phi}{2}} \cos \frac{\theta}{2} |0\rangle + e^{i\frac{\phi}{2}} \sin \frac{\theta}{2} |1\rangle.$$

$$(\because \phi := \beta - \alpha).$$

(Q) What observable has this state as eigenstate?

## Notation

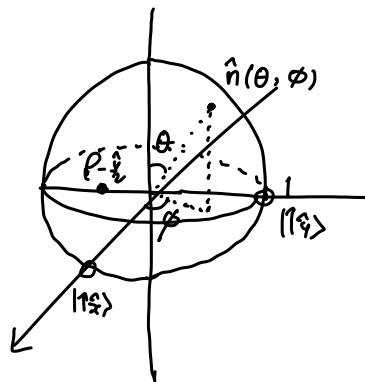
$$\text{For } \alpha = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \in \mathbb{R}^3,$$

$$a \cdot \sigma := \sum_{i=1}^3 a_i \sigma_i = \begin{pmatrix} a_3 & a_1 - i a_2 \\ a_1 + i a_2 & -a_3 \end{pmatrix}$$

not usual scalar prod.

$$\text{Let } \hat{n} = \hat{n}(\theta, \phi) = \begin{pmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{pmatrix}$$

$$\hat{n} \cdot \sigma = \begin{pmatrix} \cos\theta & e^{-i\phi} \sin\theta \\ e^{i\phi} \sin\theta & -\cos\theta \end{pmatrix}$$



: observable 'spin in direction  $\hat{n}$ '

$$\hat{n} \cdot \sigma |\psi(\theta, \phi)\rangle = |\psi(\theta, \phi)\rangle.$$

$\therefore \hat{n} \cdot \sigma$ : observable that has qubit state  $|\psi(\theta, \phi)\rangle$  as an eigenstate w. eigenval. 1.

2. Mixed state.

$$\rho = \begin{pmatrix} a & b \\ c & d \end{pmatrix}; \text{ density operator}$$

$$\Rightarrow 1. \rho^* = \rho \Rightarrow \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \begin{matrix} a, d \in \mathbb{R} \\ b = \bar{c} \end{matrix}$$

$$2. \text{tr}(\rho) = 1 \Rightarrow a + d = 1$$

for  $x_3 \in \mathbb{R}$ ,

$$3. \rho \geq 0$$

$$\begin{cases} a = \frac{1+x_3}{2} \\ d = \frac{1-x_3}{2} \\ \begin{cases} x_1 := 2\operatorname{Re}(c) \\ x_2 := 2\operatorname{Im}(c) \end{cases} \end{cases}$$

$\Rightarrow$  For  $\mathbb{X} = (x_1, x_2, x_3)$ ,

$$\rho_{\mathbb{X}} = \frac{1}{2} \begin{pmatrix} 1+x_3 & x_1 - ix_2 \\ x_1 + ix_2 & 1-x_3 \end{pmatrix} = \frac{1}{2} (\mathbb{1} + \mathbb{X} \cdot \sigma)$$

$$\downarrow \det(\rho_{\mathbb{X}} - \lambda \mathbb{1}) = \begin{vmatrix} \frac{1+x_3}{2} - \lambda & x_1 - ix_2 \\ x_1 + ix_2 & \frac{1-x_3}{2} - \lambda \end{vmatrix} = 0$$

$$\Leftrightarrow \lambda = \frac{1 \pm |\mathbb{X}|}{2} \geq 0$$

$$1 - |\mathbb{X}| \geq 0.$$

$$\therefore |\mathbb{X}| \leq 1, \quad \text{i.e. } \mathbb{X} \in \mathcal{B}_{\mathbb{R}^3}^1 \quad \text{"Bloch representation"}$$

• All edge pts  $|\mathbb{X}|=1$  correspond to pure states.

Pf). Claim) For  $a, b \in \mathbb{R}^3$ ,  $(a \cdot \sigma)(b \cdot \sigma) = (a \cdot b) \mathbb{1} + (a \times b) \cdot \sigma$

$$\begin{aligned} \text{Pf of claim). } (a \cdot \sigma)(b \cdot \sigma) &= \sum_{i,j=1}^3 a_i b_j \sigma_i \sigma_j \\ &= \sum_{i,j=1}^3 a_i b_j (\delta_{ij} \mathbb{1} + i \epsilon_{ijk} \sigma_k) \\ &= \sum_{i=1}^3 a_i b_i \mathbb{1} + i \left( a_1 b_2 \epsilon_{123} \sigma_3 + a_2 b_1 \epsilon_{213} \sigma_3 \right. \\ &\quad \left. + a_1 b_3 \epsilon_{132} \sigma_2 + a_3 b_1 \epsilon_{312} \sigma_2 \right. \\ &\quad \left. + a_2 b_3 \epsilon_{231} \sigma_1 + a_3 b_2 \epsilon_{321} \sigma_1 \right) \end{aligned}$$

$$+ a_2 b_3 \cancel{\varepsilon_{231}} \sigma_1 + a_3 b_2 \cancel{\varepsilon_{321}} \sigma_1$$

$$= (a \cdot b) \mathbb{1} + i (a \times b) \cdot \sigma$$

$$\rho_x^2 = \rho_x = \frac{1}{2} (\mathbb{1} + x \cdot \sigma)$$

$$\Leftrightarrow \frac{1}{4} (\mathbb{1} + x \cdot \sigma)^2 = \frac{1}{2} (\mathbb{1} + x \cdot \sigma)$$

$$\Leftrightarrow \mathbb{1} + \cancel{2x \cdot \sigma} + \underbrace{(x \cdot \sigma)(x \cdot \sigma)}_{(x \cdot x) \mathbb{1} + (x \times x) \cdot \sigma} = 2\mathbb{1} + \cancel{2x \cdot \sigma}$$

$$\Leftrightarrow (|x|^2 - 1) \mathbb{1} = 0$$

$$\Leftrightarrow |x| = 1.$$

\* Unitary operators on  $\mathbb{H}$ .

• For  $A^2 = \mathbb{1}$ ,  $\alpha \in \mathbb{R}$ , define

$$\begin{aligned} e^{i\alpha A} &:= \sum_{n=0}^{\infty} \frac{(i\alpha)^n}{n!} A^n \\ &= \sum_{k=0}^{\infty} \frac{(i\alpha)^{2k}}{(2k)!} A^{2k} + \sum_{j=0}^{\infty} \frac{(i\alpha)^{2j+1}}{(2j+1)!} A^{2j+1} \\ &\quad \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \alpha^{2k}}{(2k)!}}_{\cos \alpha} \underbrace{\mathbb{1}}_{\mathbb{1}} + \underbrace{\sum_{j=0}^{\infty} \frac{(-1)^j \alpha^{2j+1}}{(2j+1)!}}_{i \sin \alpha} \underbrace{A}_{A} \end{aligned}$$

$$= \cos \alpha \mathbb{1} + i \sin \alpha A.$$

- Given  $\hat{n} \in S^1_{\mathbb{R}^3}$ ,  $\alpha \in \mathbb{R}$ , the rotation around  $\hat{n}$  by angle  $\alpha$  on  $\mathcal{H}$  (i.e. spin rotation) is

$$D_{\hat{n}}(\alpha) := e^{-i \frac{\alpha}{2} \hat{n} \cdot \sigma}$$

$$= \cos\left(-\frac{\alpha}{2}\right) \mathbb{1} + i \sin\left(-\frac{\alpha}{2}\right) \hat{n} \cdot \sigma$$

$$\left( \because (\hat{n} \cdot \sigma)(\hat{n} \cdot \sigma) = (\hat{n} \cdot \hat{n}) \mathbb{1} + (\hat{n} \times \hat{n}) \cdot \sigma \right.$$

$$= \mathbb{1} \quad \left. \right)$$

$$= \cos \frac{\alpha}{2} \mathbb{1} - i \sin \frac{\alpha}{2} \hat{n} \cdot \sigma.$$

$$D_{\hat{n}}(\alpha)^* = \cos \frac{\alpha}{2} \mathbb{1} + i \sin \frac{\alpha}{2} \hat{n} \cdot \sigma.$$

$$\left( \because \hat{n} \cdot \sigma = \sum_{i=1}^3 n_i \sigma_i, \quad \sigma_i \text{ was self-adjoint} \right)$$

$$D_{\hat{n}}(\alpha) D_{\hat{n}}(\alpha)^* = \mathbb{1}$$

$$\Rightarrow D_{\hat{n}}(\alpha) \in U(\mathcal{H}).$$

\* Some expressions of  $U \in U(\mathcal{H})$ .

1.  $\exists \alpha, \beta, \gamma, \delta \in \mathbb{R}$  st  $U = e^{i\alpha} D_{\frac{\gamma}{2}}(\beta) D_{\frac{\gamma}{2}}(\gamma) D_{\frac{\gamma}{2}}(\delta)$ .

$\nwarrow \nearrow$   
 spin rotation around  
 $y$ -axis /  $z$ -axis

2.  $\exists A, B, C$ : operators,  $\alpha \in \mathbb{R}$  st  $\begin{cases} U = e^{i\alpha} A \sigma_x B \sigma_x C \\ ABC = \mathbb{1} \end{cases}$

Sketch of Pf). By 1,  $U = e^{i\alpha} D_{\frac{\gamma}{2}}(\beta) D_{\frac{\gamma}{2}}(\gamma) D_{\frac{\gamma}{2}}(\delta)$ .

Set  $A := D_{\frac{\gamma}{2}}(\beta) D_{\frac{\gamma}{2}}(\frac{\gamma}{2})$

$B := D_{\frac{\gamma}{2}}(-\frac{\gamma}{2}) D_{\frac{\gamma}{2}}(-\frac{\delta+\beta}{2})$

$C := D_{\frac{\gamma}{2}}(-\frac{\delta-\beta}{2}) \rightarrow D_{\frac{\gamma}{2}}(-\beta)$

$\Rightarrow ABC = \mathbb{1}$ .

$\Rightarrow e^{i\alpha} A \sigma_x B \sigma_x C = \dots = U$

3.  $\exists \hat{n} \in S_{\mathbb{R}^3}'$ ,  $\alpha, \frac{\gamma}{2} \in \mathbb{R}$  st.  $U = e^{i\alpha} D_{\hat{n}}(\frac{\gamma}{2})$

4.  $U = z_0 \mathbb{1} + \mathbf{z} \cdot \boldsymbol{\sigma} = \sum_{\alpha=0}^3 \frac{z_\alpha}{2} \sigma_\alpha$  for some  $z_0 \in \mathbb{C}, \mathbf{z} \in \mathbb{C}^3$

st  $|z_0|^2 + |\mathbf{z}|^2 = 1$

Pf) By 3,  $U = e^{i\alpha} D_{\hat{n}}(\frac{\gamma}{2})$ .

$= e^{i\alpha} \left( \cos \frac{\gamma}{2} \mathbb{1} - i \sin \frac{\gamma}{2} \hat{n} \cdot \boldsymbol{\sigma} \right)$

$= \underbrace{e^{i\alpha} \cos \frac{\gamma}{2} \mathbb{1}} - i \underbrace{e^{i\alpha} \sin \frac{\gamma}{2} \hat{n} \cdot \boldsymbol{\sigma}}$

$$\begin{array}{c} \text{!} \\ \vdots \\ z_0 \end{array} \quad \begin{array}{c} \text{!} \\ \vdots \\ z \end{array}$$

$$= \sum_{\alpha=0}^{\infty} z_{\alpha} \sigma_{\alpha}$$

$$|z_0|^2 + |z|^2 = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \underbrace{|\hat{n}|^2}_{=1} = 1$$

Ex. Hadamard transformation.

$$H := \frac{\sigma_x + \sigma_z}{\sqrt{2}} : |H\rangle \longrightarrow |H\rangle.$$

$$\Rightarrow H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{in the basis } \{|0\rangle, |1\rangle\}.$$

$$\begin{cases} H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ H|x_j\rangle = \frac{|0\rangle + e^{i\pi x_j} |1\rangle}{\sqrt{2}} \quad (x_j \in \{0, 1\}). \end{cases}$$

$$H = e^{i\frac{\pi}{2}} D_{\frac{1}{2}}(0) D_{\frac{1}{2}}\left(\frac{\pi}{2}\right) D_{\frac{1}{2}}(-\pi).$$

### < Ch3. Tensor products & composite systems >

#### Definition

let  $H^A, H^B$ : Hilbert spaces

$|\varphi\rangle \in H^A$ ,  $|\psi\rangle \in H^B$

Define  $|\varphi\rangle \otimes |\psi\rangle : H^A \times H^B \longrightarrow \mathbb{C}$   
 $(\xi, \eta) \longmapsto \langle \xi | \varphi \rangle \langle \eta | \psi \rangle$

$\rightarrow$  anti-linear in  $\xi$  &  $\eta$

( $\because f := |\varphi\rangle \otimes |\psi\rangle$ ,

$$\begin{aligned} f(a\xi + b\xi', \eta) &= \langle a\xi + b\xi' | \varphi \rangle \langle \eta | \psi \rangle \\ &= \overline{a} f(\xi, \eta) + \overline{b} f(\xi', \eta) \end{aligned}$$

Similar for  $\eta$ .

$$\boxed{H^A \otimes H^B} := \{ \psi : H^A \times H^B \rightarrow \mathbb{C} \mid \text{anti-linear \& anti } \eta, \exists f \}$$

$\rightarrow \mathbb{C}$ -vector space

$$(\psi_1 + \psi_2)(\xi, \eta) = \psi_1(\xi, \eta) + \psi_2(\xi, \eta)$$

$$(a\psi)(\xi, \eta) = a\psi(\xi, \eta)$$

$$\text{identity: } 0, \text{ inv } -\psi$$

\* Some expressions of  $|\psi\rangle \in U(H)$

## Properties

let  $|\psi\rangle \in H^A$ ,  $|\psi\rangle \in H^B$ ,  $a, b \in \mathbb{C}$

$$1. (a|\psi\rangle) \otimes |\psi\rangle = |\psi\rangle \otimes (a|\psi\rangle) = a(|\psi\rangle \otimes |\psi\rangle)$$

$$\begin{aligned} \therefore (a|\psi\rangle) \otimes |\psi\rangle & \stackrel{(5.7)}{=} \langle \xi | a|\psi\rangle \langle \eta | \psi \rangle \\ &= \underline{a \langle \xi | \psi \rangle \langle \eta | \psi \rangle} \\ &= \langle \xi | \psi \rangle \langle \eta | a|\psi\rangle \end{aligned}$$

$$2. (a+b)|\psi\rangle \otimes |\psi\rangle = a|\psi\rangle \otimes |\psi\rangle + b|\psi\rangle \otimes |\psi\rangle$$

$$3. (|\psi_1\rangle + |\psi_2\rangle) \otimes |\psi\rangle = |\psi_1\rangle \otimes |\psi\rangle + |\psi_2\rangle \otimes |\psi\rangle$$

$$4. |\psi\rangle \otimes (|\psi_1\rangle + |\psi_2\rangle) = |\psi\rangle \otimes |\psi_1\rangle + |\psi\rangle \otimes |\psi_2\rangle.$$

## Notation

$$|\psi \otimes \psi\rangle := |\psi\rangle \otimes |\psi\rangle$$

\*  $H^A \otimes H^B$  : Hilbert space

$$1. \text{ For } |\psi_1\rangle, |\psi_2\rangle \in H^A, |\psi_1\rangle, |\psi_2\rangle \in H^B,$$

$$|\psi_i\rangle \otimes |\psi_i\rangle \in H^A \otimes H^B.$$

$$\text{Define } \langle \psi_1 \otimes \psi_1 | \psi_2 \otimes \psi_2 \rangle = \langle \psi_1 | \psi_2 \rangle \langle \psi_1 | \psi_2 \rangle.$$

$$2. \text{ Find basis of } H^A \otimes H^B.$$

let  $\{|\psi_i\rangle\}$ ,  $\{|\psi_j\rangle\}$  : ONB of  $H^A$  &  $H^B$  resp.

Consider  $\{ |e_a \otimes f_b\rangle \} \subset H^A \otimes H^B$ .

$$1) \langle e_a \otimes f_b | e_{a'} \otimes f_{b'} \rangle = \langle e_a | e_{a'} \rangle \langle f_b | f_{b'} \rangle = \delta_{aa'} \delta_{bb'}$$

$\therefore$  orthonormal

$$2) \text{ let } \Psi \in H^A \otimes H^B.$$

$$\begin{aligned} \Psi(\xi, \eta) &= \underline{\Psi} \left( \frac{1}{a} |e_a\rangle \langle e_a| \xi \rangle, \frac{1}{b} |f_b\rangle \langle f_b| \eta \rangle \right) \\ &= \sum_{a,b} \underline{\Psi}(|e_a\rangle, |f_b\rangle) \langle \xi | e_a \rangle \langle \eta | f_b \rangle \\ &= \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle (\xi, \eta) \end{aligned}$$

$$\Psi = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

$\therefore$  generating set.

$$3) \text{ Suppose } \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle = 0 \text{ where } \Psi_{ab} \in \mathbb{C}$$

$$\Rightarrow \forall (\xi, \eta) \in H^A \times H^B,$$

$$\begin{aligned} \sum_{a,b} \Psi_{ab} (|e_a \otimes f_b\rangle) (\xi, \eta) &= \sum_{a,b} \Psi_{ab} \langle \xi | e_a \rangle \langle \eta | f_b \rangle \\ &= 0. \end{aligned}$$

$$\text{Choose } \xi = |e_{a'}\rangle, \eta = |f_{b'}\rangle$$

$$\begin{aligned} \sum_{a,b} \Psi_{ab} \langle e_{a'} | e_a \rangle \langle f_{b'} | f_b \rangle &= \sum_{a,b} \Psi_{ab} \delta_{aa'} \delta_{bb'} \\ &= \Psi_{a'b'} = 0. \end{aligned}$$

$\therefore$  l. indep.

$\therefore \{ |e_a \otimes f_b\rangle \}$  : ONB of  $H^A \otimes H^B$ .

3. For  $|\Psi\rangle, |\Phi\rangle \in H^A \otimes H^B$ , where

$$|\Psi\rangle = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

$$|\Phi\rangle = \sum_{a',b'} \Phi_{a'b'} |e_{a'} \otimes f_{b'}\rangle,$$

$$\begin{aligned} \text{Define } \langle \Psi | \Phi \rangle &:= \sum_{a,b} \sum_{a',b'} \overline{\Psi_{ab}} \Phi_{a'b'} \underbrace{\langle e_a \otimes f_b | e_{a'} \otimes f_{b'} \rangle}_{\substack{\delta_{aa'} \delta_{bb'}}} \\ &= \sum_{a,b} \overline{\Psi_{ab}} \Phi_{ab} \end{aligned}$$